Practice entity-relationship transformations to ER models

Create Relational model schemas (NOT an ERD) that allows storing each of the data described below. Use a separate relational schema for each case. Mark each primary key by underlining it, and each foreign key with its REFERENCES. Assume that all entities, unless explicitly specified, have at least one ID attribute and a “name” or “description” attribute:

1. **A department employs many people. A person is employed by, at most, one department.**

   DEPARTMENT(DepartmentID, Name)

   PERSON(PersonID, Name, DepartmentID (FK)) WHERE {DepartmentID} REFERENCES DEPARTMENT

2. **A manager manages, at most, one department. A department is managed by, at most, one manager.**

   MANAGER(ManagerID, Name)

   DEPARTMENT(DepartmentID, Name, ManagerID(FK) (UNIQUE)) WHERE {ManagerID} REFERENCES MANAGER

3. **An author may write many books. A book may be written by many authors.**

   AUTHOR(AuthorID, Name)

   BOOK(BookID, Name, Description)

   WRITES(AuthorID, BookID) WHERE {AuthorID} REFERENCES AUTHOR AND WHERE {BookID} REFERENCES BOOK

4. **A lecturer teaches many courses. A course is taught by exactly one lecturer.**

   LECTURER(LecturerID, Name)

   COURSE(CourseID, Name, LecturerID(FK) (NOT NULL)) WHERE {LecturerID} REFERENCES LECTURER

5. **A purchase order may be for many products. A product may appear on many purchase orders.**

   ORDER(OrderID, Description)

   PRODUCT(ProductID, Name)

   BUY(OrderID(FK), ProductID(FK)) WHERE {OrderID} REFERENCES ORDER AND WHERE {ProductID} REFERENCES PRODUCT

6. **A customer may submit many orders. An order is for exactly one customer.**

   CUSTOMER(CustomerID, Name)

   ORDER(OrderID, Description, CustomerID(FK) (NOT NULL)) WHERE {CustomerID} REFERENCES CUSTOMER

7. **An employee may supervise many other employees. Each employee is supervised by, at most, one other employee. An employee cannot supervise themselves.**

   EMPLOYEE(EmployeeID, Name, SupervisorID(FK)) WHERE {SupervisorID} REFERENCES EMPLOYEE

8. **A student may join many clubs. A club may have many students. Each membership must store the date the student joined the club.**

   STUDENT(StudentID, Name)

   CLUB (ClubID, Name)

   JOINS(ClubID(FK), StudentID(FK), JoinDate (NOT NULL)) WHERE {StudentID} REFERENCES STUDENT AND WHERE {ClubID} REFERENCES CLUB

9. **A doctor may prescribe medications to patients. A patient may receive prescriptions from many doctors, and a medication may be prescribed to many patients by many doctors. Each prescription must store the prescription date and dosage.**

   DOCTOR(DoctorID, Name)

   PATIENT(PatientID, Name)

   MEDICATION(MedicationID, Name)

   PRESCRIPTION(DoctorID(FK), PatientID(FK), MedicationID(FK), PrescriptionDate (NOT NULL), Dosage(NOT NULL)) WHERE {DoctorID} REFERENCES DOCTOR AND WHERE {PatientID} REFERENCES PATIENT AND WHERE {MedicationID} REFERENCES MEDICATION

10. **Each country has exactly one capital city, and each capital city is the capital of exactly one country.**

    CAPITAL (CapitalID, Name)

    COUNTRY(CountryID, Name, CapitalID(FK) (UNIQUE) (NOT NULL)) WHERE {CapitalID} REFERENCES CAPITAL

Note/Hints:

- If mandatory participation is not stated, consider the relation optional.
- “may” or “can” represent possibilities.
- “must” or “exactly” represent obligation.
- If a case may be represented by both key propagation or a new table for the relationship, write down both.